

PYTHON PRACTICAL ASSIGNMENT

Rule-Based Conversational Chatbot

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ASSIGNMENT	30-Minute Student Task

1. Aim

To develop a simple rule-based conversational chatbot in Python that can continuously interact with a user, recognize different messages, provide appropriate responses, and allow the user to exit the conversation.

2. Program

```
# Rule-Based Conversational Chatbot
# Student: Marusu Lavanya
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print("=== SIMPLE PYTHON CHATBOT ===")

name = input("Enter your name: ").strip()
print("Hello", name + "!")
print("You can ask me about Python, my name, or the weather.")
print("Type 'bye' or 'exit' to stop.")

while True:
    user_message = input("You: ").lower().strip()

    if user_message == "bye" or user_message == "exit":
        print("Bot: Goodbye", name + "! See you again.")
        break

    elif "hello" in user_message or "hi" in user_message:
        print("Bot: Hi", name + "! Nice to chat with you.")

    elif "python" in user_message:
        print("Bot: Python is a popular programming language used in many areas.")

    elif "how are you" in user_message:
        print("Bot: I am fine and ready to answer your questions.")

    elif "your name" in user_message:
        print("Bot: I am a rule-based chatbot created using Python.")

    elif "weather" in user_message:
        print("Bot: I cannot access live weather information right now.")

    else:
        print("Bot: I did not understand that. Please try another question.")
```

3. Logic Used

Python Concept	Purpose in the chatbot
input()	Accepts the student's name and conversation messages.
.lower().strip()	Normalizes each message before checking it.

Python Concept	Purpose in the chatbot
while True:	Keeps the conversation running repeatedly.
if / elif / else	Selects the appropriate response.
in operator	Recognizes greeting variations such as 'hello bot'.
break	Terminates the chatbot after 'bye' or 'exit'.

4. Sample Output

```

=== SIMPLE PYTHON CHATBOT ===
Enter your name: Lavanya
Hello Lavanya!
You can ask me about Python, my name, or the weather.
Type 'bye' or 'exit' to stop.

You: hello bot
Bot: Hi Lavanya! Nice to chat with you.

You: what is python
Bot: Python is a popular programming language used in many areas.

You: how are you
Bot: I am fine and ready to answer your questions.

You: random message
Bot: I did not understand that. Please try another question.

You: exit
Bot: Goodbye Lavanya! See you again.

```

5. Requirement Verification

✓	User's name is requested before the main conversation.
✓	The chatbot uses while True for continuous interaction.
✓	Messages are collected using input().
✓	Input is cleaned with .lower().strip().
✓	hello and hi greetings are recognized.
✓	Multiple topics/questions receive responses.
✓	if / elif / else decision making is used.
✓	bye and exit terminate the loop using break.
✓	The user's name appears in chatbot responses.
✓	Unknown messages receive a useful else response.

6. Conclusion

The Python chatbot satisfies the task requirements by following the INPUT → PROCESS → RESPOND → REPEAT → EXIT flow. It demonstrates basic Python concepts including strings, input handling, loops, conditional statements, the in operator, and break.